Name:

Student ID:

Q1: A convolutional layer in a CNN has 32 filters, each with a kernel size of 3 × 3, applied to an input with 16 channels. Assuming each filter has a bias term, calculate the total number of trainable parameters in this layer.

Q2: A fully connected (linear) layer in a neural network has 128 input neurons and 64 output neurons. Each neuron in the output layer has a bias term, calculate the total number of trainable parameters in this layer.

Q3: True or False: In a Convolutional Neural Network (CNN), the deeper layers (further from input) are responsible for extracting fundamental visual patterns such as edges, textures, and simple geometric shapes. These layers work at a granular level, ensuring that basic visual structures are well understood before passing the information to higher layers. On the other hand, the shallower layers (closer to input) take this information and transform it into more complex, abstract representations, allowing the network to recognize high-level concepts like object categories and scene understanding. This hierarchical structure ensures that the network can gradually build up to meaningful representations from the simplest building blocks.

Q4:A Vision Transformer (ViT) takes an input image of size 224 × 224 with 3 channels (RGB) and splits it into fixed-size patches before processing them as tokens in a transformer model.

(a) If the patch size is 16 × 16, how many patches (tokens) will the ViT generate?

(b) What is the dimensionality of each patch (token) before flattening?

(c) After flattening each patch into a token, what is the final shape of the input sequence fed into the Transformer? *(Assume no additional positional embedding for now.)*

Q5: What is mode collapse in GANs and why does it occur?

Q6: What is the intuition behind denoising diffusion probabilistic models (DDPM)?
Describe the forward and reverse processes.

Q7: Why do diffusion models gradually add noise instead of adding noise all at once? And what happens when a very fast noise schedule (very few timesteps) is applied, what happens to the diffusion model performance and why?

Q8: What is mode collapse in GANs

Q9: Why is it beneficial to mask contiguous patches (SimMIM) rather than random individual patches (MAE)?

Mark Scheme

Question	Answer - Total = 25 points
Q1: A convolutional layer in a CNN has 32 filters, each with a kernel size of 3 × 3, applied to an input with 16 channels. Assuming each filter has a bias term, calculate the total number of trainable parameters in this layer.	**Total - 3 points** Option1: Correct answer 3 points without showing your working Option2: Correct formula 1 point Correct substitution 1 point Correct answer 1 point Option3: Almost correct answer, the only thing wrong is your bias term 2 points Total Parameters = (KW × KH ×IC + BT) × #filters = (3 × 3 × 16 + 1) × 32 = 4640 KW = Kernel Width KH = Kernel Height IC = Input Channels BT = bias term #Fitler = number of filters
Q2: A fully connected (linear) layer in a neural network has 128 input neurons and 64 output neurons. Each neuron in the output layer has a bias term.	**Total - 3 points** Option1: Correct answer 3 points without showing your working Option2: Correct formula 1 point Correct substitution 1 point Correct answer 1 point Option3: Almost correct answer, the only thing wrong is your bias term 2 points Total Parameters = (IN x ON) + BT = 128 x 64 + 64 = 8256 IN = Input Neurons ON = Output Neurons

	BT = Bias terms
Q3: True or False: In a Convolutional Neural Network (CNN), the deeper layers (further from input) are responsible for extracting fundamental visual patterns such as edges, textures, and simple geometric shapes. These layers work at a granular level, ensuring that basic visual structures are well understood before passing the information to higher layers. On the other hand, the shallower layers (closer to input) take this information and transform it into more complex, abstract representations, allowing the network to recognize high-level concepts like object categories and scene understanding. This hierarchical structure ensures that the network can gradually build up to meaningful representations from the simplest building blocks.	**Total - 1 point** Correct answer - 2 points False
Q4:A Vision Transformer (ViT) takes an input image of size 224 × 224 with 3 channels (RGB) and splits it into fixed-size patches before processing them as tokens in a transformer model. (a) If the patch size is 16 × 16, how many patches (tokens) will the ViT generate? (b) What is the dimensionality of each patch (token) before flattening? (c) After flattening each patch into a token, what is the final shape of the input sequence fed into the Transformer? *(Assume no additional positional embedding for now.*)	**(a) Total - 3 points** Option1: Correct answer 3 points without showing your working Option2: Correct width 1 point Correct height 1 point Correct answer 1 point Total = Image size/ patch size = (224 x 224) / (16x 16) H = 224/16 W = 224/16 Total = 196 **(b) Total - 3 points** Option1: Correct answer 3 points without showing your working Option2: Correct patch size 1 point Correct channel 1 point Correct answer 1 point

	Dim = patch size * channel = 16 x 16 x 3 = 768 **(c) Total - 1 point** Correct answer - 1 point Final shape: (196,768)
Q5: What is mode collapse in GANs and why does it occur?	**Total - 2 points** Mode collapse in GANs refers to the phenomenon where the generator produces a limited variety of outputs, often mapping different inputs to the same or very similar outputs, thus failing to capture the full diversity (or "modes") of the data distribution. - 1 point It occurs because the generator finds a shortcut that successfully fools the discriminator with a narrow set of outputs. Since the discriminator only provides scalar feedback (real vs. fake), it may not penalize lack of diversity, especially if the few outputs are realistic enough. - 1 point * Open to other discussions
Q6: What is the intuition behind denoising diffusion probabilistic models (DDPM)? Describe the forward and reverse processes.	**Total - 3 points** DDPMs generate data by reversing a gradual noising process. - 1 point Forward process: Adds Gaussian noise to data over many steps until it becomes pure noise. - 1 point Reverse process: A neural network learns to denoise step-by-step, recovering realistic data from noise. - 1 point
Q7: Why do diffusion models gradually add noise instead of adding noise all at once? And what happens when a very fast noise schedule (very few timesteps) is applied, what happens to the diffusion model performance and why?	**Total - 3 points** Diffusion models add noise gradually to make the noising process invertible and learnable—each small step allows the model to learn to denoise effectively. If noise is

	added all at once, the mapping from noise to data becomes too complex to learn. - 1 point With a very fast noise schedule (few timesteps), model performance drops because each step must undo too much noise, making denoising harder and less stable. - 1 point The model struggles to recover fine details, leading to blurry or poor-quality outputs. - 1 point
Q8: What is mode collapse in GANs	Same as Q5
Q9: Why is it beneficial to mask contiguous patches (SimMIM) rather than random individual patches (MAE)?	**Total - 1 point** Contiguous masking (SimMIM) helps the model learn semantic structure by reconstructing coherent regions, unlike random masking (MAE), which may rely on local cues. It encourages better contextual understanding and often leads to faster, more stable training. *Open to any other discussions